# Associations between elevated homocysteine, cognitive impairment, and reduced white matter volume in healthy old adults.

OBJECTIVES: Elevated homocysteine has emerged as a risk factor for cognitive impairment even in healthy elderly persons. Reduced brain volume and white matter hyperintensities also occur in healthy elderly as well, but the interrelationships between these have not been well studied. We report these interrelationships in non demented, relatively healthy, community-dwelling older adults from a single East Asian population. METHODS: Two hundred twenty-eight right-handed participants age 55 years and above were evaluated. Persons with medical conditions or neurological diseases other than well-controlled diabetes mellitus and hypertension were excluded. Participants underwent quantitative magnetic resonance imaging of the brain using a standardized protocol and neuropsychological evaluation. Plasma homocysteine, folate, vitamin B(12), and markers for cardiovascular risk: blood pressure, body mass index, fasting blood glucose, and lipid profile were measured. RESULTS: Elevated homocysteine was associated with reduced global cerebral volume, larger ventricles, reduced cerebral white matter volume, and lower cognitive performance in several domains. Elevated homocysteine was associated with reduced white matter volume ($\beta = -20.80$, $t = -2.9$, $df = 223$, $p = 0.004$) and lower speed of processing ($\beta = -0.38$, $t = -2.1$, $df = 223$, $p = 0.03$), even after controlling for age, gender, and education. However, the association between homocysteine and lower speed of processing disappeared after controlling for white matter volume. Elevated homocysteine was not associated with white matter hyperintensity volume or with hippocampal volume. Although homocysteine and folate levels were correlated, their effects on white matter volume were dissociated. CONCLUSION: In non demented, relatively healthy adults, elevated homocysteine is associated with lower cognitive scores and reduced cerebral white matter volume. These effects can be dissociated from those related to white matter hyperintensities or reduced folate level. 